US Visa Prediction Project

Life cycle of Machine learning Project

- Understanding the Problem Statement
- Data Collection
- Exploratory data analysis
- Data Cleaning
- Data Pre-Processing
- Model Training
- Choose best model

About

The Immigration and Nationality Act (INA) of the US permits foreign workers to come to the United States to work on either a temporary or permanent basis. The act also protects US workers against adverse impacts on working place and maintain requirements when they hire foreign workers to fill workforce shortages. The immigration programs are administered by the Office of Foreign Labor Certification (OFLC).

1) Problem statement.

- OFLC gives job certification applications for employers seeking to bring foreign workers into the United States and grants certifications.
- As In last year the count of employees were huge so OFLC needs Machine learning models to shortlist visa applicants based on their previous data.

In this project we are going to use the data given to build a Classification model:

- This model is to check if Visa get approved or not based on the given dataset.
- This can be used to Recommend a suitable profile for the applicants for whom the visa should be certified or denied based on the certain criteria which influences the decision.

2) Data Collection.

- The Dataset is part of Office of Foreign Labor Certification (OFLC)
- The data consists of 25480 Rows and 12 Columns

https://www.kaggle.com/datasets/moro23/easyvisa-dataset

2.1 Import Data and Required Packages

Importing Pandas, Numpy, Matplotlib, Seaborn and Warings Library.

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import warnings

warnings.filterwarnings("ignore")

%matplotlib inline
```

Import the CSV Data as Pandas DataFrame

In [2]:
```
df = pd.read_csv(r"Visadataset.csv")
```

Show Top 5 Records

In [3]:
```
df.head()
```

Out[3]:

	case_id	continent	education_of_employee	has_job_experience	requires_job_training	no
0	EZYV01	Asia	High School	N	N	
1	EZYV02	Asia	Master's	Y	N	
2	EZYV03	Asia	Bachelor's	N	Y	
3	EZYV04	Asia	Bachelor's	N	N	
4	EZYV05	Africa	Master's	Y	N	

Shape of the dataset

In [4]:
```
df.shape
```

Out[4]:
```
(25480, 12)
```

Summary of the dataset

In [5]:
```
# Display summary statistics for a dataframe
df.describe()
```

Out[5]:

	no_of_employees	yr_of_estab	prevailing_wage
count	25480.000000	25480.000000	25480.000000
mean	5667.043210	1979.409929	74455.814592
std	22877.928848	42.366929	52815.942327
min	-26.000000	1800.000000	2.136700
25%	1022.000000	1976.000000	34015.480000
50%	2109.000000	1997.000000	70308.210000
75%	3504.000000	2005.000000	107735.512500
max	602069.000000	2016.000000	319210.270000

Check Datatypes in the dataset

In [6]:
```
# Check Null and Dtypes
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25480 entries, 0 to 25479
Data columns (total 12 columns):
 #   Column                 Non-Null Count  Dtype  
---  ------                 --------------  -----  
 0   case_id                25480 non-null  object 
 1   continent              25480 non-null  object 
 2   education_of_employee  25480 non-null  object 
 3   has_job_experience     25480 non-null  object 
 4   requires_job_training  25480 non-null  object 
 5   no_of_employees        25480 non-null  int64  
 6   yr_of_estab            25480 non-null  int64  
 7   region_of_employment   25480 non-null  object 
 8   prevailing_wage        25480 non-null  float64
 9   unit_of_wage           25480 non-null  object 
 10  full_time_position     25480 non-null  object 
 11  case_status            25480 non-null  object 
dtypes: float64(1), int64(2), object(9)
memory usage: 2.3+ MB
```

3. EXPLORING DATA

In [7]:
```
# define numerical & categorical columns
numeric_features = [feature for feature in df.columns if df[feature].dtype != 'O']
categorical_features = [feature for feature in df.columns if df[feature].dtype == '

# print columns
print('We have {} numerical features : {}'.format(len(numeric_features), numeric_fe
print('\nWe have {} categorical features : {}'.format(len(categorical_features), ca
```

```
We have 3 numerical features : ['no_of_employees', 'yr_of_estab', 'prevailing_wage']

We have 9 categorical features : ['case_id', 'continent', 'education_of_employee',
'has_job_experience', 'requires_job_training', 'region_of_employment', 'unit_of_wag
e', 'full_time_position', 'case_status']
```

In [8]:
```
# proportion of count data on categorical columns
for col in categorical_features:
    print(df[col].value_counts(normalize=True) * 100)
    print('----------------------------')
```

```
case_id
EZYV01       0.003925
EZYV16995    0.003925
EZYV16993    0.003925
EZYV16992    0.003925
EZYV16991    0.003925
               ...
EZYV8492     0.003925
EZYV8491     0.003925
EZYV8490     0.003925
EZYV8489     0.003925
EZYV25480    0.003925
Name: proportion, Length: 25480, dtype: float64
----------------------------
continent
Asia             66.173469
Europe           14.646782
North America    12.919937
South America     3.343799
Africa            2.162480
Oceania           0.753532
Name: proportion, dtype: float64
----------------------------
education_of_employee
Bachelor's     40.164835
Master's       37.810047
High School    13.422292
Doctorate       8.602826
Name: proportion, dtype: float64
----------------------------
has_job_experience
Y    58.092622
N    41.907378
Name: proportion, dtype: float64
----------------------------
requires_job_training
N    88.402669
Y    11.597331
Name: proportion, dtype: float64
----------------------------
region_of_employment
Northeast    28.237834
South        27.539246
West         25.847724
Midwest      16.903454
Island        1.471743
Name: proportion, dtype: float64
----------------------------
unit_of_wage
Year     90.117739
Hour      8.465463
Week      1.067504
Month     0.349294
Name: proportion, dtype: float64
----------------------------
full_time_position
```

```
Y    89.375981
N    10.624019
Name: proportion, dtype: float64
---------------------------
case_status
Certified    66.789639
Denied       33.210361
Name: proportion, dtype: float64
---------------------------
```

Insights

- `case_id` have unique vlaues for each column which can be dropped as it it of no importance
- `continent` column is highly biased towards asia. hence we can combine other categories to form a single category.
- `unit_of_wage` seems to be an important column as most of them are yearly contracts.

Univariate Analysis

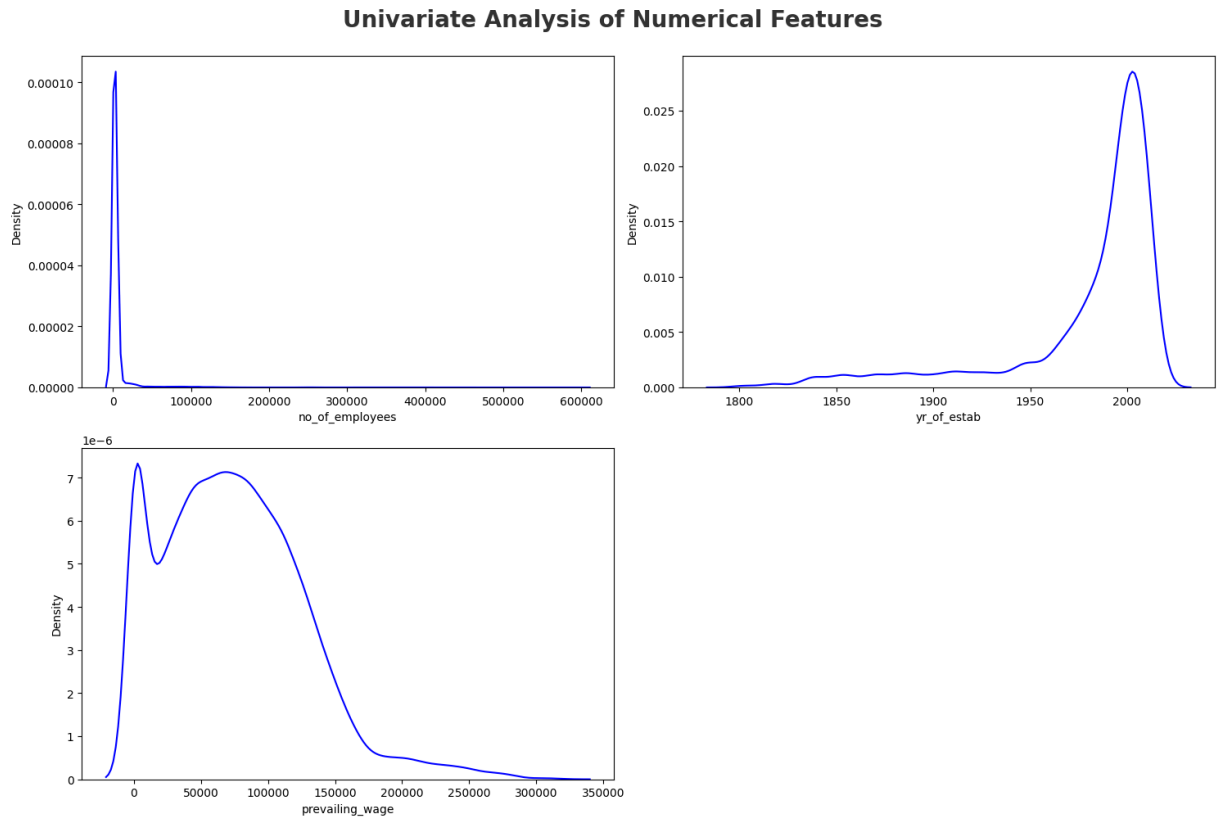
- The term univariate analysis refers to the analysis of one variable prefix “uni” means “one.” The purpose of univariate analysis is to understand the distribution of values for a single variable.

Other Type of Analysis are

- **Bivariate Analysis:** The analysis of two variables.
- **Multivariate Analysis:** The analysis of two or more variables.

Numerical Features

In [9]:
```
plt.figure(figsize=(15, 10))
plt.suptitle('Univariate Analysis of Numerical Features', fontsize=20, fontweight='

for i in range(0, len(numeric_features)):
    plt.subplot(2, 2, i+1)
    sns.kdeplot(x=df[numeric_features[i]], color='blue')
    plt.xlabel(numeric_features[i])
    plt.tight_layout()

# save plot
# plt.savefig('./images/Univariate_Num.png')
```

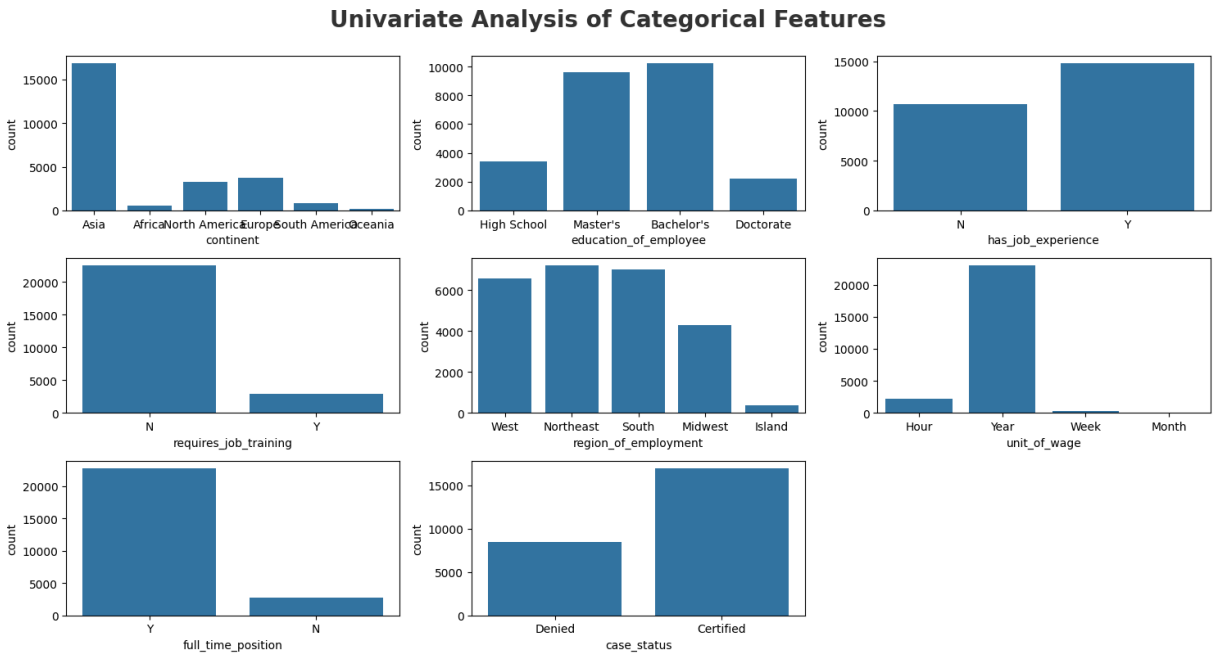


Insights

- `no_of_employees` column looks more skewed but it may be because of outlies.
- `yr_of_estab` is a discrete feature.
- `privailing-wage` is a right-skewed distribution.

Categorical Features

In [10]:
```
# categorical columns
categorical_features.remove('case_id')
plt.figure(figsize=(15, 8))
plt.suptitle('Univariate Analysis of Categorical Features', fontsize=20, fontweight

for i in range(0, len(categorical_features)):
    plt.subplot(3, 3, i+1)
    sns.countplot(x=df[categorical_features[i]])
    plt.xlabel(categorical_features[i])
    plt.tight_layout()

# save plot
# plt.savefig('./images/Univariate_Cat.png')
```



Insights

- `requires_job_training`, `unit_of_wage`, `full_time_position` and `continents` coulmns have signle category dominating.
- In rest of the columns are balanced.

Multivariate Analysis

- Multivariate analysis is the analysis of more than one variable.

In [11]:
```
discrete_features=[feature for feature in numeric_features if len(df[feature].uniqu

continuous_features=[feature for feature in numeric_features if len(df[feature].uni

print('We have {} discrete features : {}'.format(len(discrete_features), discrete_f
print('\nWe have {} continuous_features : {}'.format(len(continuous_features), cont
```

```
We have 0 discrete features : []

We have 3 continuous_features : ['no_of_employees', 'yr_of_estab', 'prevailing_wag
e']
```

Check Multicollinearity in Numerical features

In [14]:
```
df.head()
```

Out[14]:

	case_id	continent	education_of_employee	has_job_experience	requires_job_training	no
0	EZYV01	Asia	High School	N	N	
1	EZYV02	Asia	Master's	Y	N	
2	EZYV03	Asia	Bachelor's	N	Y	
3	EZYV04	Asia	Bachelor's	N	N	
4	EZYV05	Africa	Master's	Y	N	

Insights

- There is no multicollinearity between any variables

Check Multicollinearity for Categorical features

- **A chi-squared test (also chi-square or χ2 test) is a statistical hypothesis test that is valid to perform when the test statistic is chi-squared distributed under the null hypothesis, specifically Pearson's chi-squared test**
- **A chi-square statistic is one way to show a relationship between two categorical variables.**
- **Here we test correlation of Categorical columns with Target column i.e case_status**

Null Hypothesis (H_0): The Feature is independent of target column (No-Correlation)

Alternative Hypothesis (H_1): The Feature and Target column are not independent (Correalted)

In [19]:

```
from scipy.stats import chi2_contingency
chi2_test = []
for feature in categorical_features:
    if chi2_contingency(pd.crosstab(df['case_status'], df[feature]))[1] < 0.05:
        chi2_test.append('Reject Null Hypothesis')
    else:
        chi2_test.append('Fail to Reject Null Hypothesis')
result = pd.DataFrame(data=[categorical_features, chi2_test]).T
result.columns = ['Column', 'Hypothesis Result']
result
```

Out[19]:

	Column	Hypothesis Result
0	continent	Reject Null Hypothesis
1	education_of_employee	Reject Null Hypothesis
2	has_job_experience	Reject Null Hypothesis
3	requires_job_training	Fail to Reject Null Hypothesis
4	region_of_employment	Reject Null Hypothesis
5	unit_of_wage	Reject Null Hypothesis
6	full_time_position	Reject Null Hypothesis
7	case_status	Reject Null Hypothesis

Report

- Here `requires_job_training` fails to Rejects Null Hypothesis which means it doesn't correlate with target column.

Checking Null Values

In [20]:
```
df.isnull().sum()
```

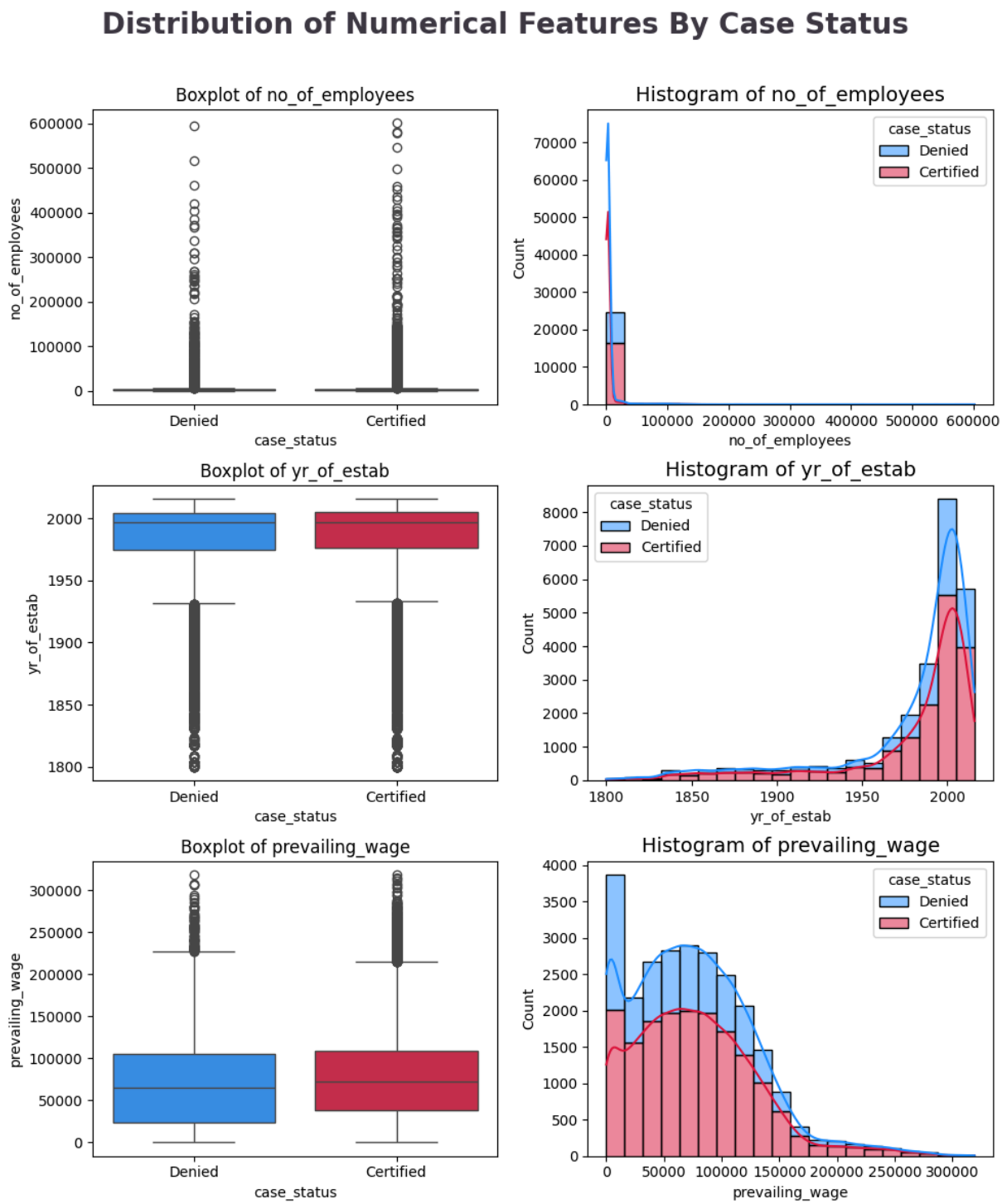
Out[20]:
```
case_id                  0
continent                0
education_of_employee    0
has_job_experience       0
requires_job_training    0
no_of_employees          0
yr_of_estab              0
region_of_employment     0
prevailing_wage          0
unit_of_wage             0
full_time_position       0
case_status              0
dtype: int64
```

In [21]:
```
continues_features=[feature for feature in numeric_features if len(df[feature].uniq
print('Num of continues features :',continues_features)
```

```
Num of continues features : ['no_of_employees', 'yr_of_estab', 'prevailing_wage']
```

In [22]:
```
clr1 = ['#1E90FF', '#DC143C']
fig, ax = plt.subplots(3, 2, figsize=(10,12))
fig.suptitle('Distribution of Numerical Features By Case Status', color='#3C3744',
             fontsize=20, fontweight='bold', ha='center')
for i, col in enumerate(continues_features):
    sns.boxplot(data=df, x='case_status', y=col,  palette=clr1, ax=ax[i,0])
    ax[i,0].set_title(f'Boxplot of {col}', fontsize=12)
    sns.histplot(data=df, x=col, hue='case_status', bins=20, kde=True,
                 multiple='stack', palette=clr1, ax=ax[i,1])
```

```
    ax[i,1].set_title(f'Histogram of {col}', fontsize=14)
fig.tight_layout()
fig.subplots_adjust(top=0.90)
# plt.savefig('images/multivariate_num.png')
```



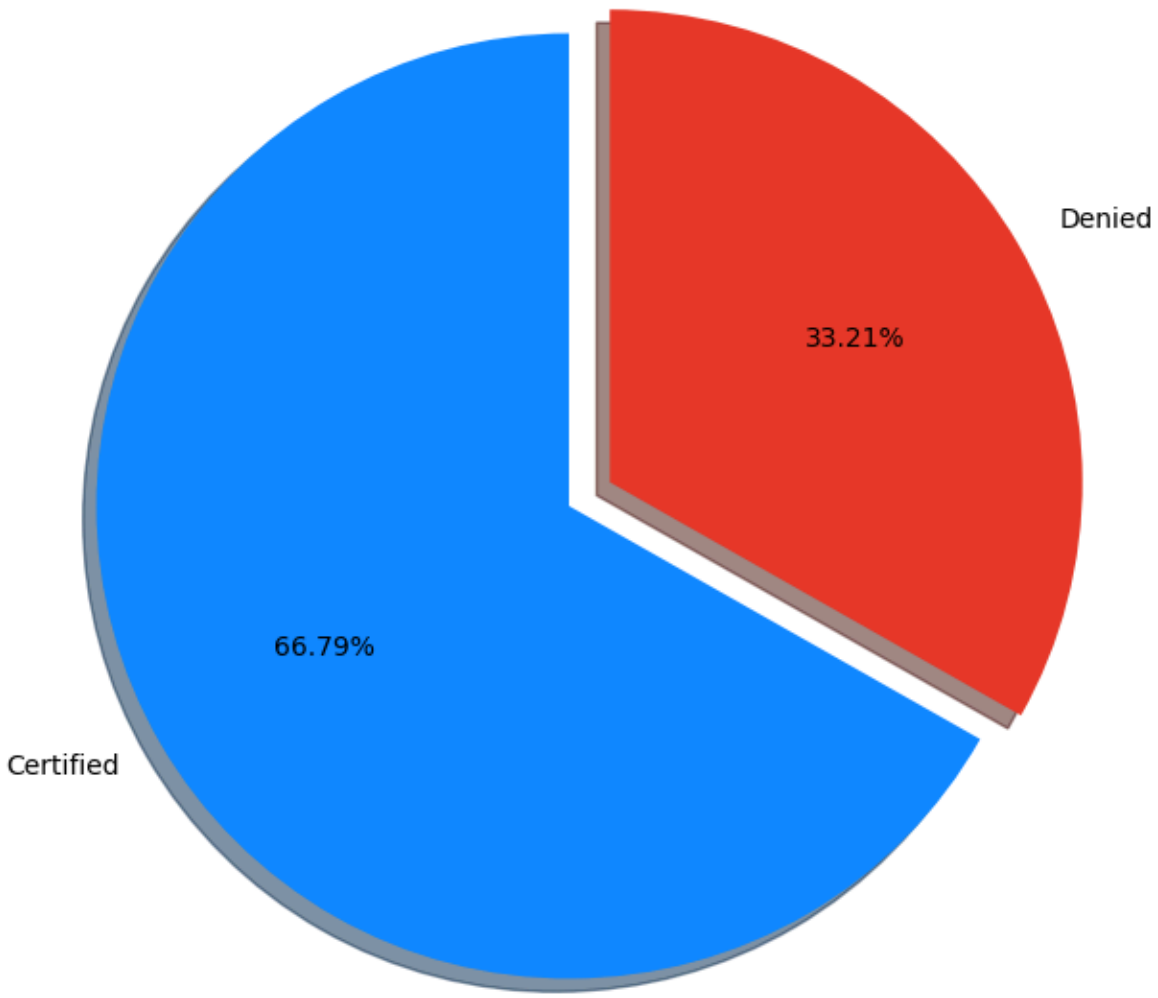
Initial Analysis Report

- **No of Employees has many outliers which can be Handled in Feature Engineering and `no_of_employees` is Right Skewed.**
- **`yr_of_estab` is left skewed and some outliers below the lower bound of Box plot.**
- **`prevailing_wage` is right skewed with outliers above upper bound of box plot.**
- **There are No missing values in the dataset.**
- **The `case_id` column can be deleted because each row has unique values.**
- **The `case_status` column is the target to predict.**
- **In the Categorical column, features can be made Binary numerical in feature Encoding**

4. Visualization

4.1 Visualize the Target Feature

In [23]:
```
percentage = df.case_status.value_counts(normalize=True)*100
labels = ["Certified","Denied"]

# Plot PieChart with Ploty library
fig, ax = plt.subplots(figsize =(15, 8))
explode = (0, 0.1)
colors = ['#1188ff','#e63a2a']
ax.pie(percentage, labels = labels, startangle = 90,
       autopct='%1.2f%%',explode=explode, shadow=True, colors=colors)
plt.show()
```



- From the chart it is seen that the Target Variable is Imbalanced

What is imbalanced data?

Imbalanced data are types of data where the target class has an uneven distribution of observations, i.e Here Denied value has more count than the Certified value of the dataset.

4.2 Does applicant Continent has any impact on Visa status ?

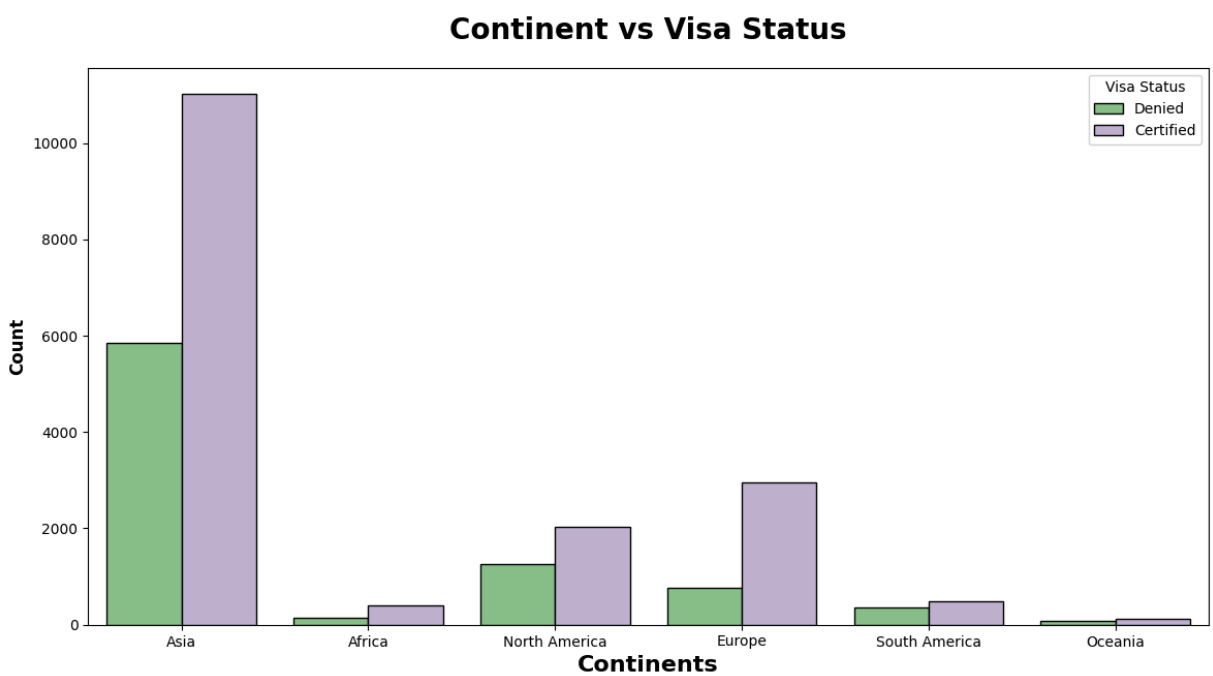
In [24]:
```
#group data by continent and their count of case_status
df.groupby('continent')['case_status'].value_counts(normalize=True).to_frame()*100
```

Out[24]:

continent	case_status	proportion
Africa	**Certified**	72.050817
	Denied	27.949183
Asia	**Certified**	65.310480
	Denied	34.689520
Europe	**Certified**	79.233655
	Denied	20.766345
North America	**Certified**	61.877278
	Denied	38.122722
Oceania	**Certified**	63.541667
	Denied	36.458333
South America	**Certified**	57.863850
	Denied	42.136150

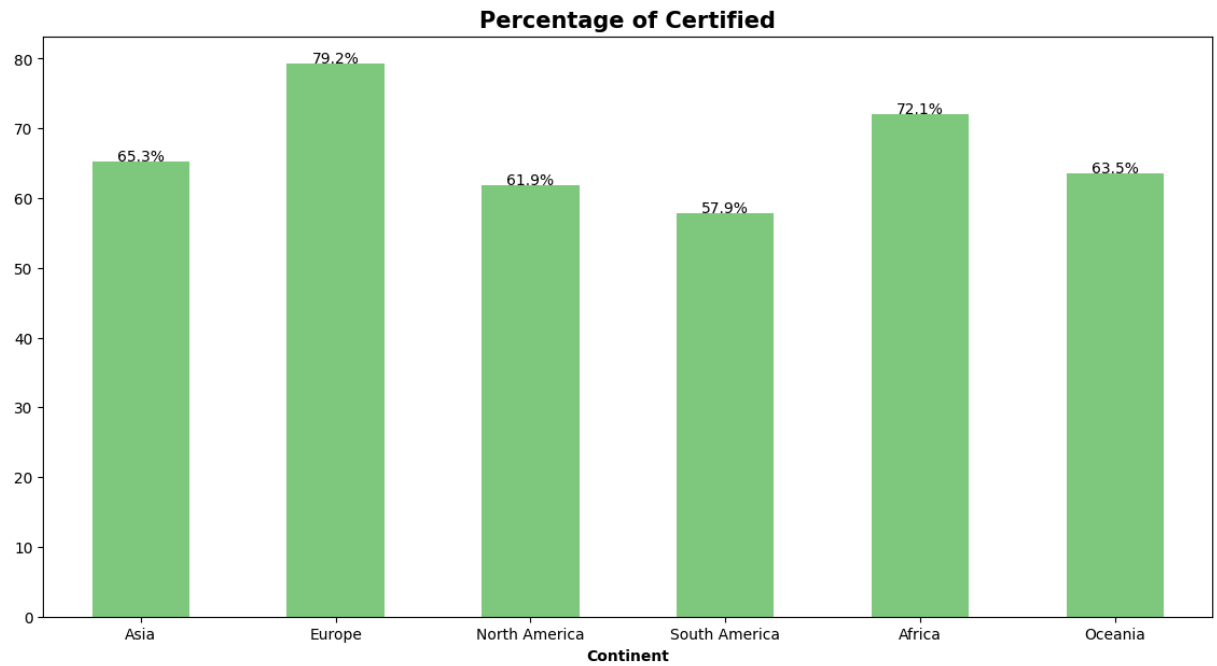
In [25]:

```
plt.subplots(figsize=(14,7))
sns.countplot(x="continent",hue="case_status", data=df, ec = "black",palette="Accen
plt.title("Continent vs Visa Status", weight="bold",fontsize=20, pad=20)
plt.ylabel("Count", weight="bold", fontsize=12)
plt.xlabel("Continents", weight="bold", fontsize=16)
plt.legend(title="Visa Status", fancybox=True)
plt.show()
```



In [26]:
```
df2 = df.copy()
```

In [27]:
```
plt.figure(figsize=[14,7])

(100*df2[df2["case_status"].isin(['Certified'])]['continent'].value_counts()/df2['c
    kind='bar',stacked=True , colormap='Accent')
plt.title("Percentage of Certified" , fontsize = 15, fontweight ='bold'  )
order1 = (100*df2[df2["case_status"].isin(['Certified'])]['continent'].value_counts
for n in range(order1.shape[0]):
    count = order1[n]
    strt='{:0.1f}%'.format(count)
    plt.text(n,count+0.1,strt,ha='center')

plt.xlabel('Continent' , fontweight ='bold')
plt.xticks(rotation=0)
plt.show()
```



Report:

- As per the Chart Asia applicants applied more than other continents.
- 43% of Certified applications are from Asia.
- This is followed by Europe with 11% of Certified applications.
- Highest chance of getting certified if you are from `Europe` and followed by `Africa`

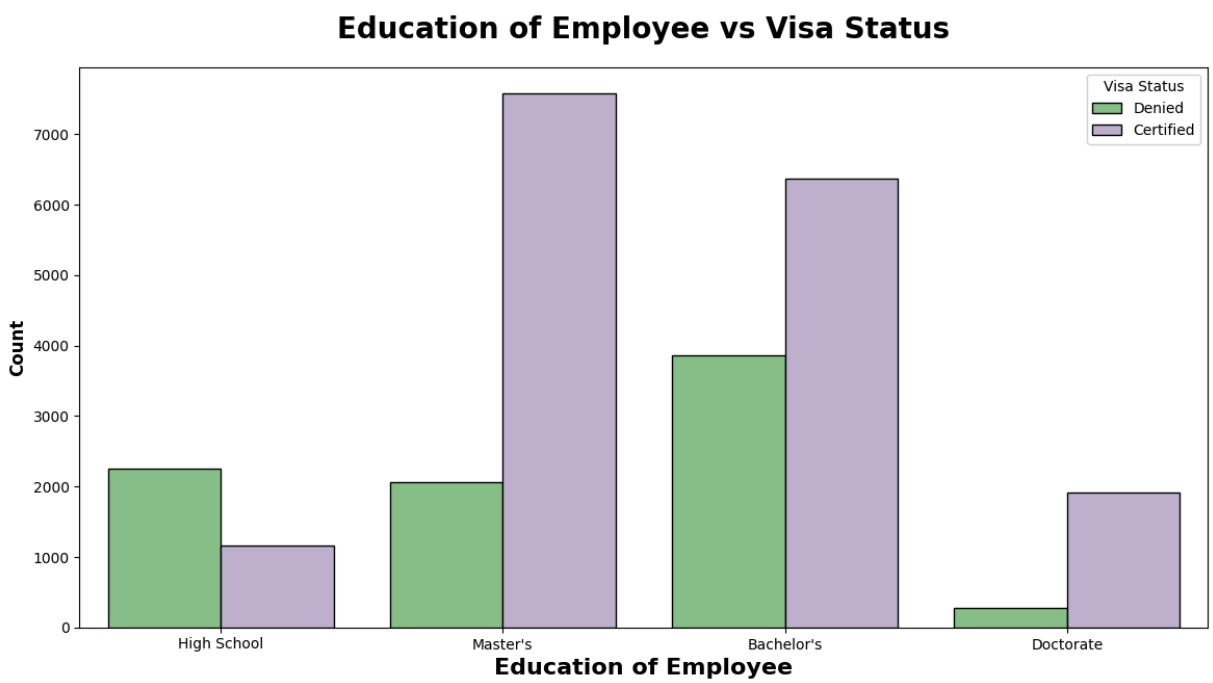
4.3 Does applicant education has any impact on Visa status ?

In [28]:
```
#group data by Education and their count of case_status
df.groupby('education_of_employee')['case_status'].value_counts(normalize=True).to_
```

Out[28]:

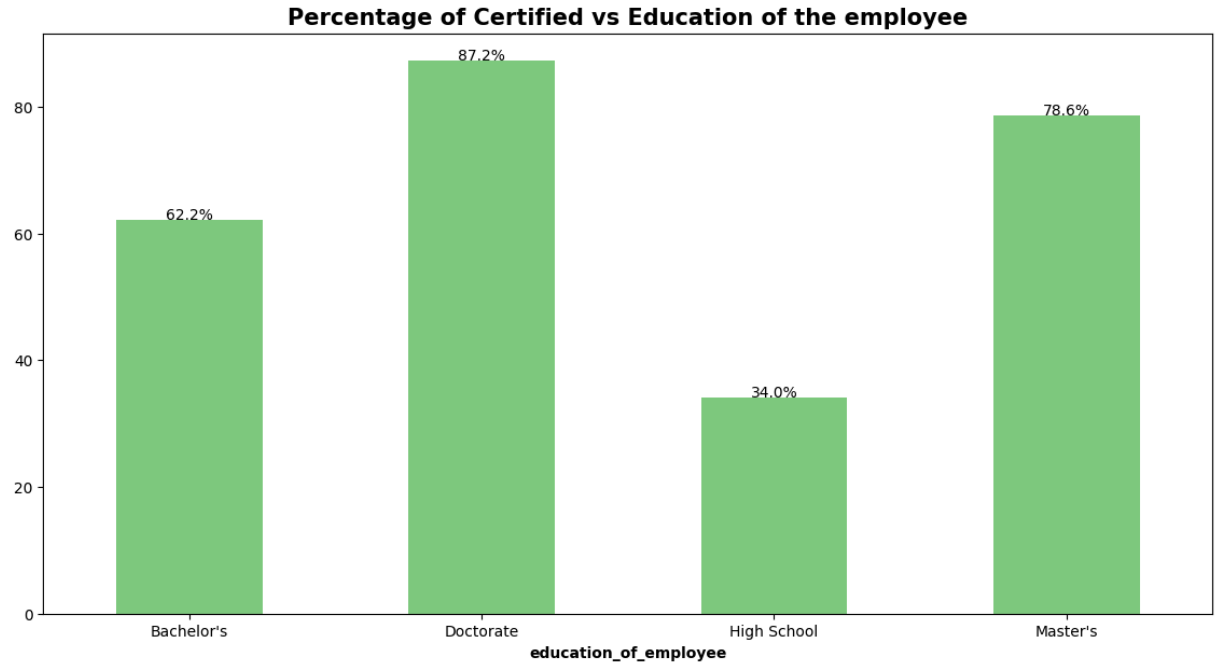
education_of_employee	case_status	proportion
Bachelor's	Certified	62.214188
	Denied	37.785812
Doctorate	Certified	87.226277
	Denied	12.773723
High School	Denied	65.964912
	Certified	34.035088
Master's	Certified	78.627777
	Denied	21.372223

In [29]:
```
plt.subplots(figsize=(14,7))
sns.countplot(x="education_of_employee",hue="case_status", data=df, ec = "black",pa
plt.title("Education of Employee vs Visa Status", weight="bold",fontsize=20, pad=20
plt.ylabel("Count", weight="bold", fontsize=12)
plt.xlabel("Education of Employee", weight="bold", fontsize=16)
plt.legend(title="Visa Status", fancybox=True)
plt.show()
```



In [30]:
```
plt.figure(figsize=[14,7])

(100*df2[df2["case_status"].isin(['Certified'])]['education_of_employee'].value_cou
    kind='bar',stacked=True , colormap='Accent')
plt.title("Percentage of Certified vs Education of the employee" , fontsize = 15, f
order1 = (100*df2[df2["case_status"].isin(['Certified'])]['education_of_employee'].
for n in range(order1.shape[0]):
    count = order1[n]
```

```
    strt='{:0.1f}%'.format(count)
    plt.text(n,count+0.1,strt,ha='center')

plt.xlabel('education_of_employee' , fontweight ='bold')
plt.xticks(rotation=0)
plt.show()
```



Insights

- education status has high impact
- `Doctorate` and `Master's` graduates have higher cange of being accepted then the others.

4.4 Does applicant's previous work experience has any impact on Visa status ?

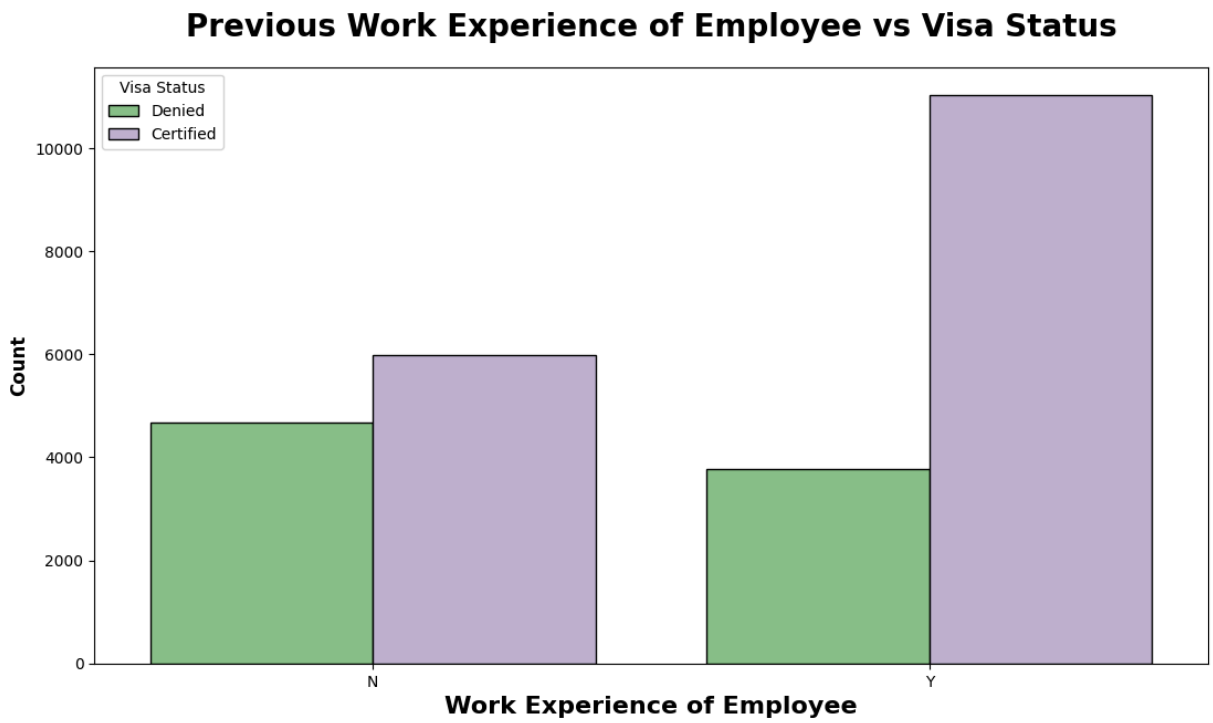
In [31]:
```
#group data by has_job_experience and count case_status
df.groupby('has_job_experience')['case_status'].value_counts(normalize=True).to_fra
```

Out[31]:

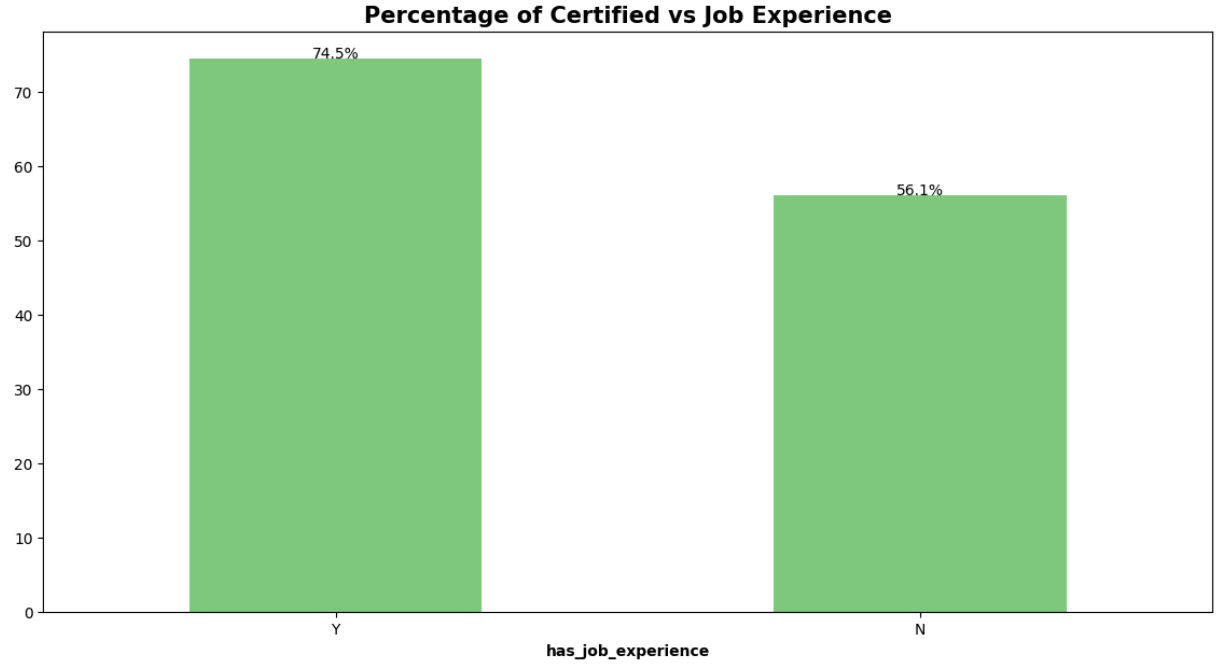
has_job_experience	case_status	proportion
N	Certified	56.134108
	Denied	43.865892
Y	Certified	74.476422
	Denied	25.523578

In [32]:
```
plt.subplots(figsize=(13,7))
sns.countplot(x="has_job_experience",hue="case_status", data=df, ec= "black",palett
```

```
plt.title("Previous Work Experience of Employee vs Visa Status", weight="bold",font
plt.ylabel("Count", weight="bold", fontsize=12)
plt.xlabel("Work Experience of Employee", weight="bold", fontsize=16)
plt.legend(title="Visa Status", fancybox=True)
plt.show()
```



In [33]:
```
plt.figure(figsize=[14,7])

(100*df2[df2["case_status"].isin(['Certified'])]['has_job_experience'].value_counts
    kind='bar',stacked=True , colormap='Accent')
plt.title("Percentage of Certified vs Job Experience" , fontsize = 15, fontweight =
order1 = (100*df2[df2["case_status"].isin(['Certified'])]['has_job_experience'].val
for n in range(order1.shape[0]):
    count = order1[n]
    strt='{:0.1f}%'.format(count)
    plt.text(n,count+0.1,strt,ha='center')

plt.xlabel('has_job_experience' , fontweight ='bold')
plt.xticks(rotation=0)
plt.show()
```



Report

- People with previous work experience has been certified 74.5% only 25.5% applicant has been denied.
- People with No previous work experience has been certified 56% and denied 43%.
- This means work experience has effect on Visa status.
- There is a slight edge for the people with work experiences then the fresheres as expected. But the difference is not huge.

4.5 If the Employee requires job training, does it make any impact on visa status?

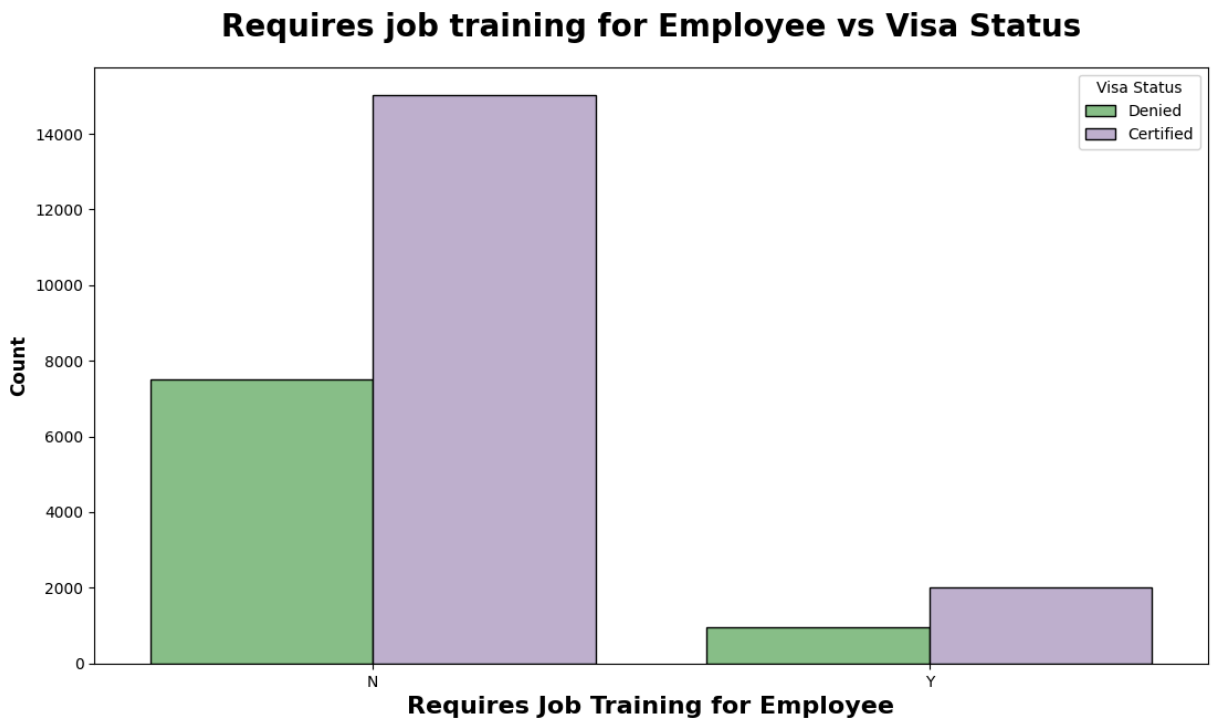
In [34]:
```
#group data by requires_job_training and count case_status
df.groupby('requires_job_training')['case_status'].value_counts(normalize=True).to_
```

Out[34]:

requires_job_training	case_status	proportion
N	Certified	66.645949
	Denied	33.354051
Y	Certified	67.884941
	Denied	32.115059

In [35]:
```
plt.subplots(figsize=(13,7))
sns.countplot(x="requires_job_training",hue="case_status", data=df, ec= "black",pal
plt.title("Requires job training for Employee vs Visa Status", weight="bold",fontsi
plt.ylabel("Count", weight="bold", fontsize=12)
plt.xlabel("Requires Job Training for Employee", weight="bold", fontsize=16)
```

```
plt.legend(title="Visa Status", fancybox=True)
plt.show()
```



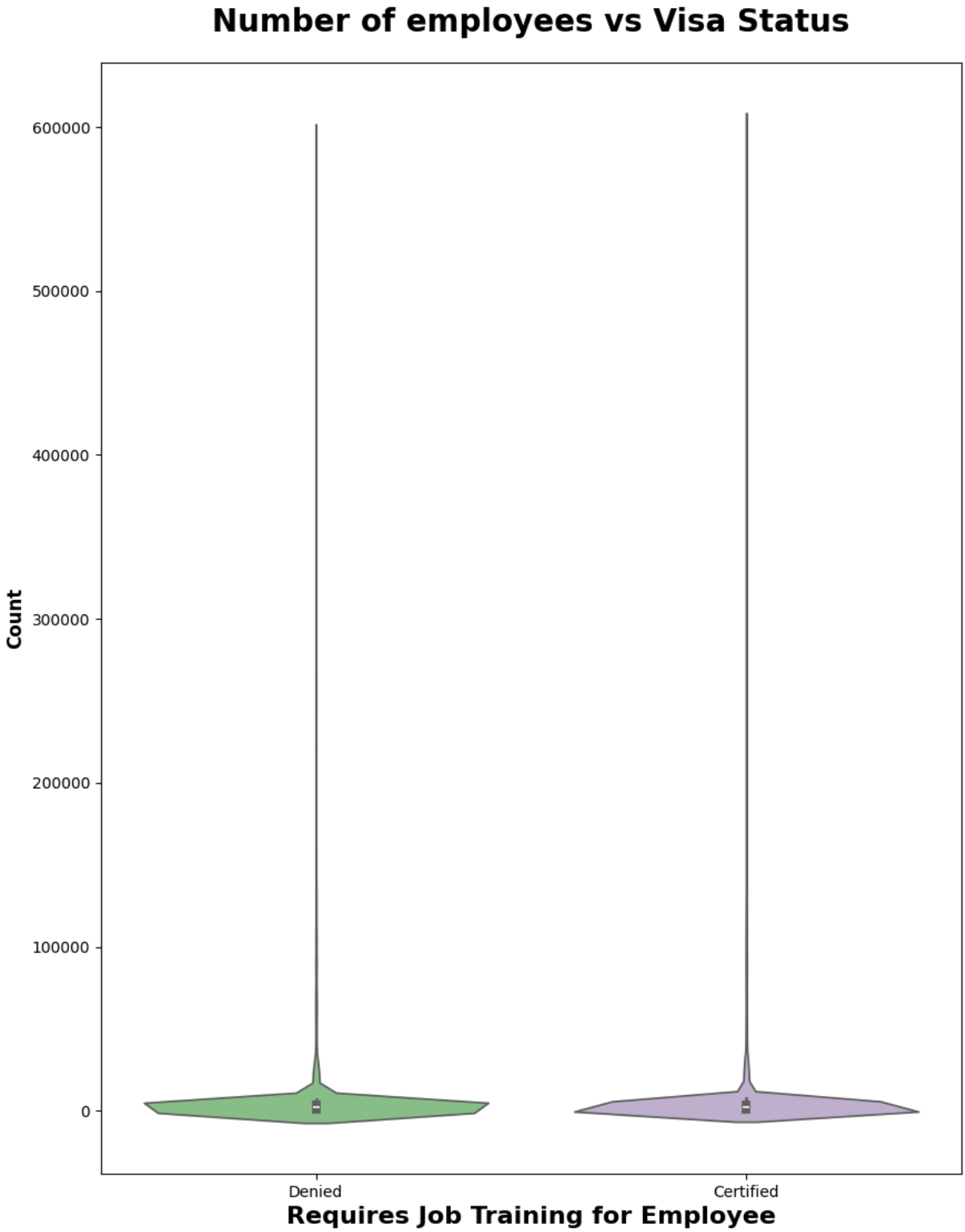
In [36]:
```
plt.figure(figsize=[14,7])

(100*df2[df2["case_status"].isin(['Certified'])]['requires_job_training'].value_cou
    kind='bar',stacked=True , colormap='Accent')
plt.title("Percentage of Certified vs Required training" , fontsize = 15, fontweigh
order1 = (100*df2[df2["case_status"].isin(['Certified'])]['requires_job_training'].
for n in range(order1.shape[0]):
    count = order1[n]
    strt='{:0.1f}%'.format(count)
    plt.text(n,count+0.1,strt,ha='center')

plt.xlabel('requires_job_training' , fontweight ='bold')
plt.xticks(rotation=0)
plt.show()
```

Report

- If employee requires job training and it doesn't have any effect on Visa status.
- 88% of applicant don't require job training.
- 63% of people who doesnt want job training got certified.
- For employees who require job training 67% of people get certified.
- As we checked in Chi-Squared test this feature doesn't have much impact on target variable, which is confirmed by above plot.

4.6 Does Number of employees of employer has any impact on Visa status?

In [37]:
```
plt.subplots(figsize=(10,13))
sns.violinplot(x="case_status",y='no_of_employees', data=df, palette="Accent")
plt.title("Number of employees vs Visa Status", weight="bold",fontsize=20, pad=20)
plt.ylabel("Count", weight="bold", fontsize=12)
plt.xlabel("Requires Job Training for Employee", weight="bold", fontsize=16)
plt.ylim()
plt.show()
```



Insights

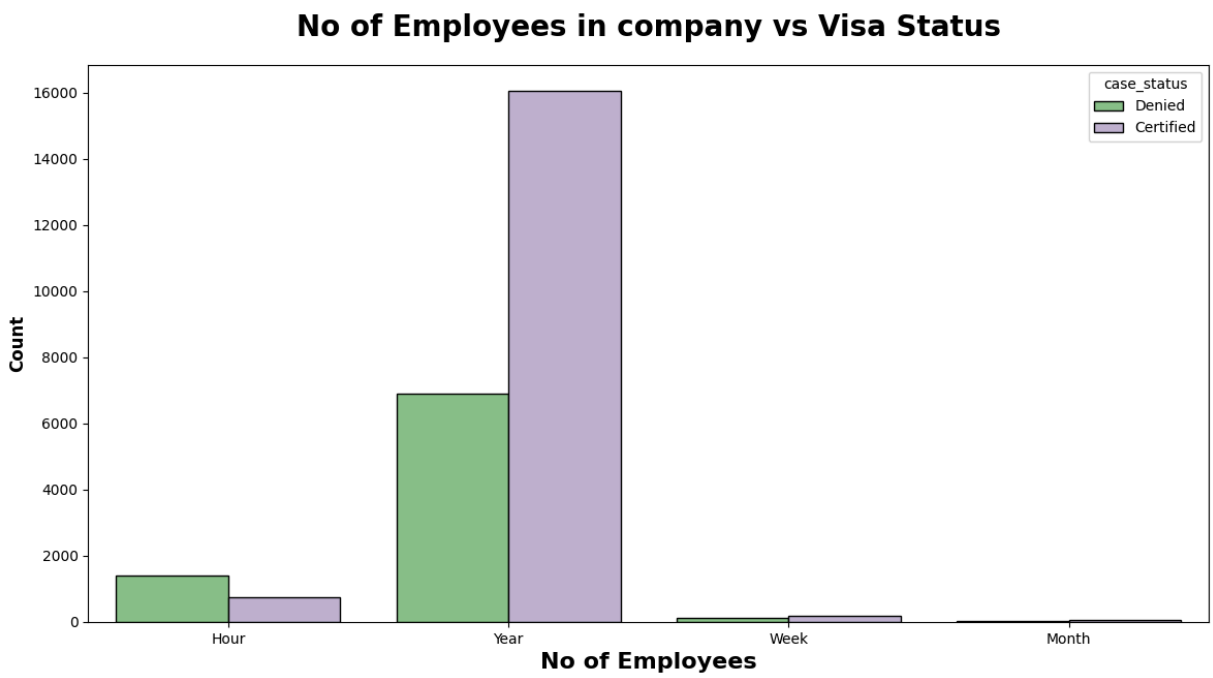
- The distrubution of both is similar. But there are outliers in both the classes which need to be handeled.

4.7 Wages and its impact on Visa status

In [38]:
```
#group data by unit_of_wage and count the case_status
df.groupby('unit_of_wage')['case_status'].value_counts(normalize=True)
```

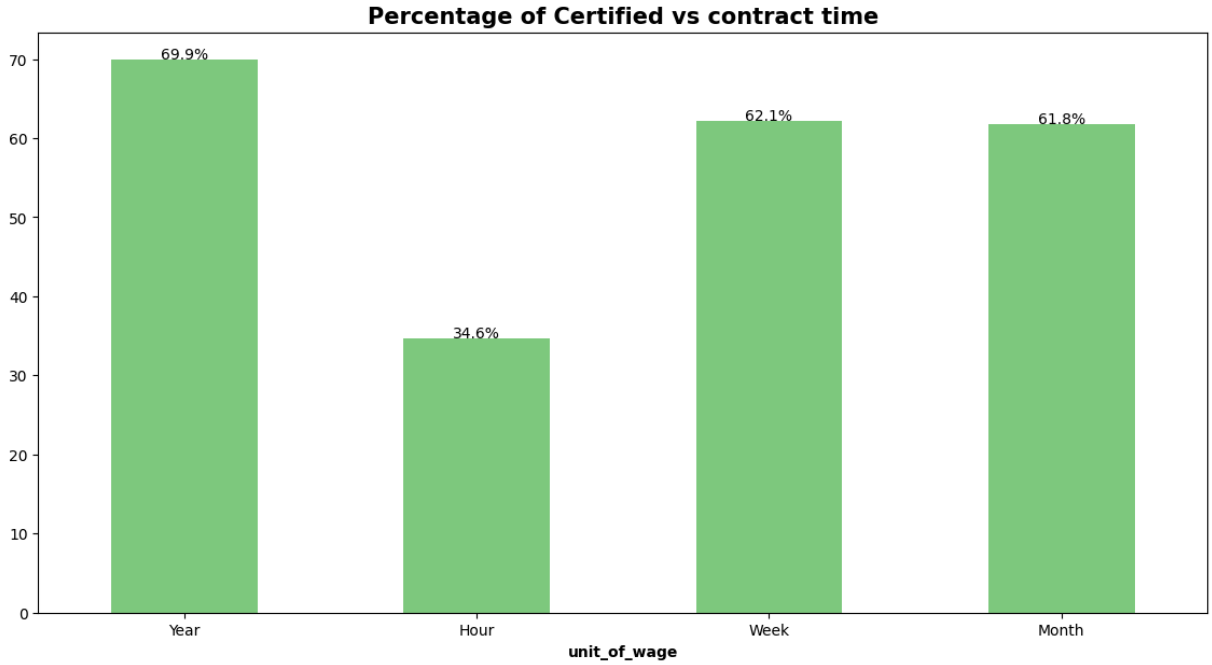
Out[38]:
```
unit_of_wage  case_status
Hour          Denied         0.653686
              Certified      0.346314
Month         Certified      0.617978
              Denied         0.382022
Week          Certified      0.621324
              Denied         0.378676
Year          Certified      0.698850
              Denied         0.301150
Name: proportion, dtype: float64
```

In [39]:
```
plt.subplots(figsize=(14,7))
sns.countplot(x="unit_of_wage",hue="case_status", data=df,ec='k', palette='Accent')
plt.title("No of Employees in company vs Visa Status", weight="bold",fontsize=20, p
plt.xlabel("No of Employees", weight="bold", fontsize=16)
plt.ylabel("Count", weight="bold", fontsize=12)
plt.show()
```



In [40]:
```
plt.figure(figsize=[14,7])

(100*df2[df2["case_status"].isin(['Certified'])]['unit_of_wage'].value_counts()/df2
    kind='bar',stacked=True , colormap='Accent')
plt.title("Percentage of Certified vs contract time" , fontsize = 15, fontweight ='
order1 = (100*df2[df2["case_status"].isin(['Certified'])]['unit_of_wage'].value_cou
for n in range(order1.shape[0]):
    count = order1[n]
    strt='{:0.1f}%'.format(count)
    plt.text(n,count+0.1,strt,ha='center')

plt.xlabel('unit_of_wage' , fontweight ='bold')
```

```
plt.xticks(rotation=0)
plt.show()
```

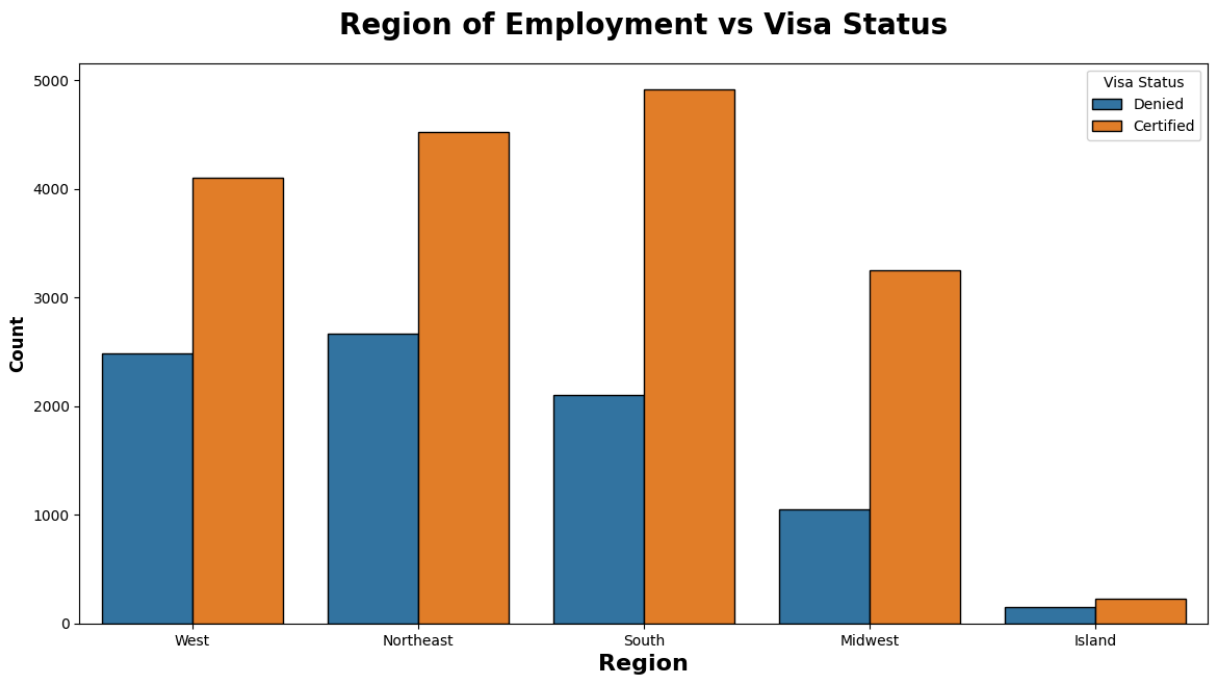


Report

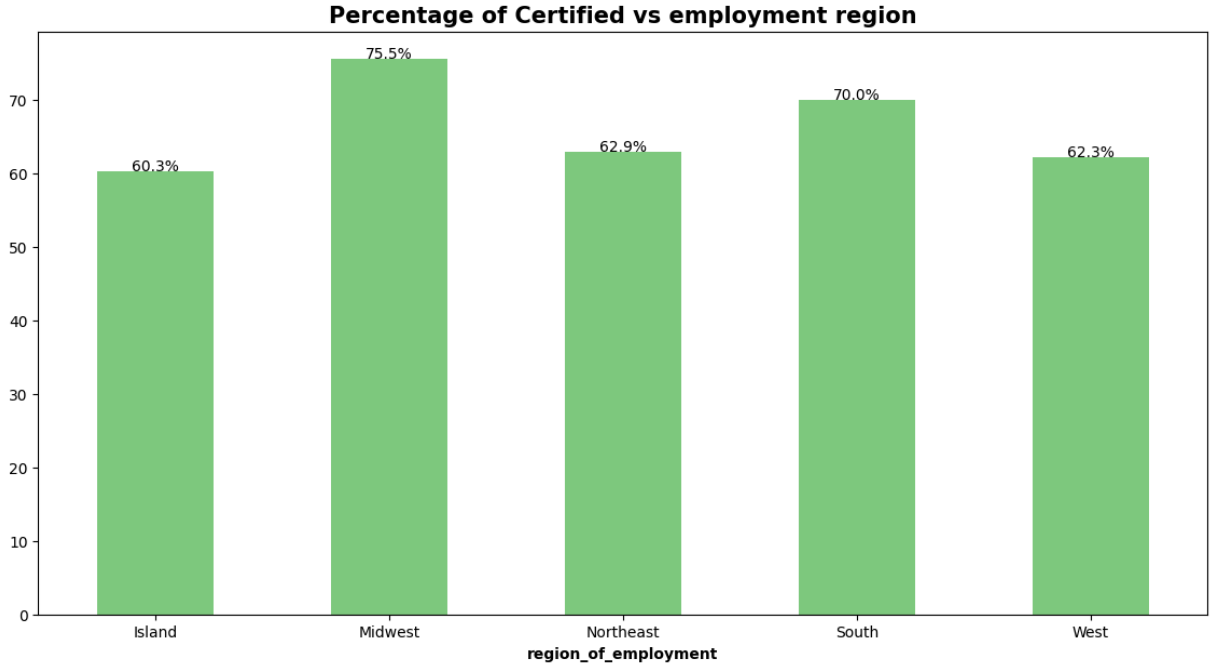
- For employees who applied for hourly pay 65% of them were denied.
- Yearly unit wage application were accepted for 69% and denied for 31% of the time.
- There is a higher chance of yearly contract than other and immediately followed by week and month.

4.7 Does Region of employment has impact on Visa status ?

In [41]:
```
plt.subplots(figsize=(14,7))
sns.countplot(x="region_of_employment",hue="case_status", data=df,ec='k')
plt.title("Region of Employment vs Visa Status", weight="bold",fontsize=20, pad=20)
plt.xlabel("Region", weight="bold", fontsize=16)
plt.ylabel("Count", weight="bold", fontsize=12)
plt.legend(title="Visa Status", fancybox=True)
plt.show()
```



In [42]:
```
plt.figure(figsize=[14,7])

(100*df2[df2["case_status"].isin(['Certified'])]['region_of_employment'].value_coun
    kind='bar',stacked=True , colormap='Accent')
plt.title("Percentage of Certified vs employment region" , fontsize = 15, fontweigh
order1 = (100*df2[df2["case_status"].isin(['Certified'])]['region_of_employment'].v
for n in range(order1.shape[0]):
    count = order1[n]
    strt='{:0.1f}%'.format(count)
    plt.text(n,count+0.1,strt,ha='center')

plt.xlabel('region_of_employment' , fontweight ='bold')
plt.xticks(rotation=0)
plt.show()
```



Report

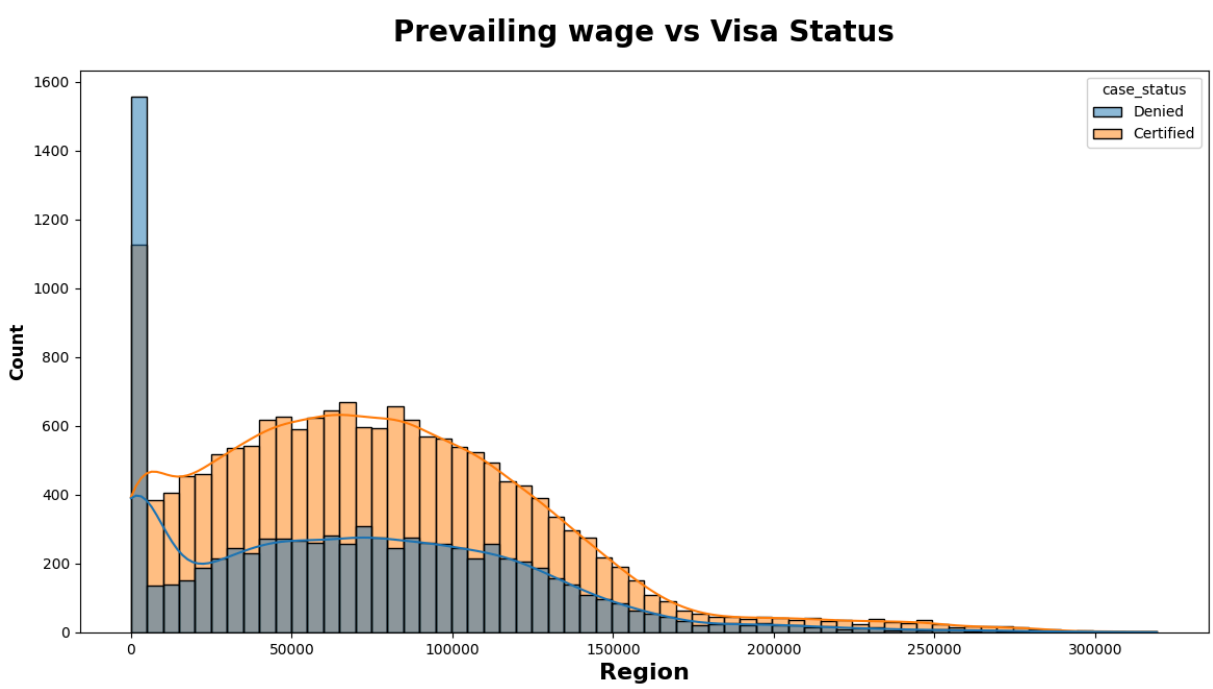
- As per chart all the region have very similar pattern of getting Visa certified and Denied.
- There is a slight edge for `midwest` followed by `South` region

4.8 Does Prevailing wage has any impact on Visa status ?

In [43]:
```
df.groupby('prevailing_wage').case_status.value_counts()
```

Out[43]:
```
prevailing_wage  case_status
2.1367           Certified      1
2.9561           Denied         1
3.0031           Denied         1
3.3188           Denied         1
3.4889           Certified      1
                                ..
311734.4900      Certified      1
314156.0600      Certified      1
315497.6000      Certified      1
318446.0500      Certified      1
319210.2700      Denied         1
Name: count, Length: 25462, dtype: int64
```

In [44]:
```
plt.subplots(figsize=(14,7))
sns.histplot(x="prevailing_wage",hue="case_status", data=df, kde=True)
plt.title("Prevailing wage vs Visa Status", weight="bold",fontsize=20, pad=20)
plt.xlabel("Region", weight="bold", fontsize=16)
plt.ylabel("Count", weight="bold", fontsize=12)
plt.show()
```



Insights

- The distribution of both the classes are same but need to handel the outliers.

4.8.1 Prevailing wage based on Education

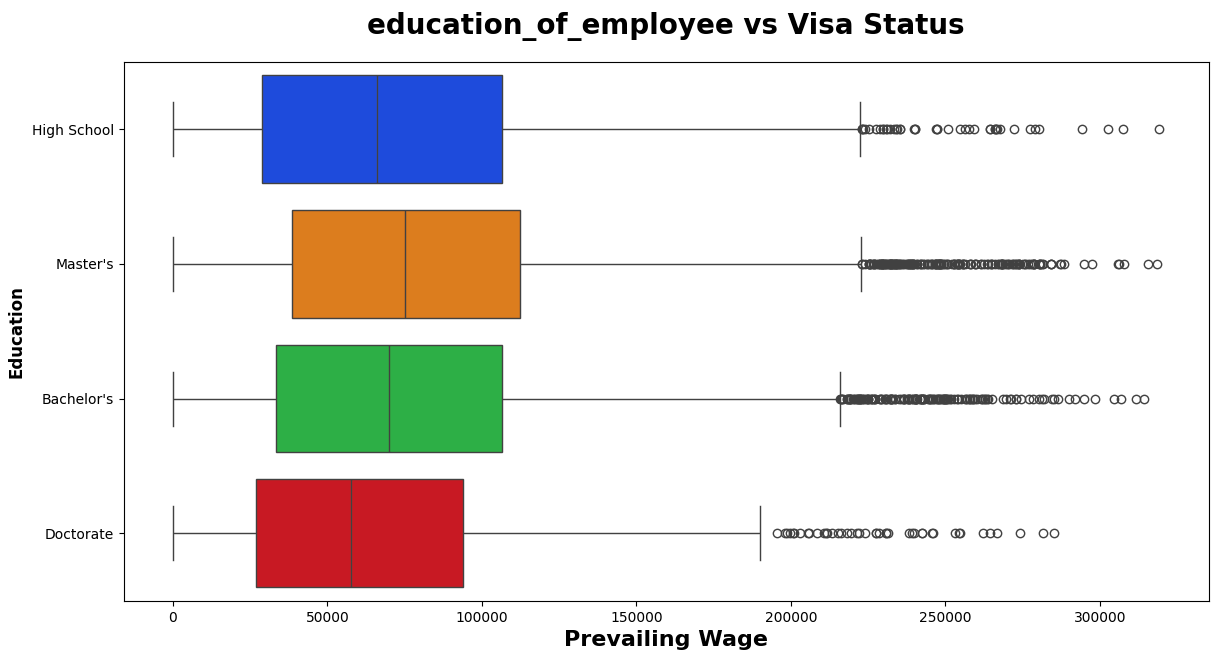
In [45]:
```
print('Average Prevailing wage based on Employee education')
df.groupby('education_of_employee')['prevailing_wage'].mean().to_frame().sort_value
```

```
Average Prevailing wage based on Employee education
```

Out[45]:

education_of_employee	prevailing_wage
Master's	78843.057843
Bachelor's	73405.443735
High School	71582.147756
Doctorate	64561.076657

In [46]:
```
plt.subplots(figsize=(14,7))
sns.boxplot(y="education_of_employee",x = "prevailing_wage", data=df,palette='brigh
plt.title("education_of_employee vs Visa Status", weight="bold",fontsize=20, pad=20
plt.xlabel("Prevailing Wage", weight="bold", fontsize=16)
plt.ylabel("Education", weight="bold", fontsize=12)
plt.show()
```



Report

- Based on the above table and charts its seen that applicant with Master's education has higher average prevailing wages.
- Master's education applicants have median salary of $78.8k, which approximately 20% higher than Doctrate's average wage, which is strange.

4.8.2 Prevailing wage based on Job experience

In [47]:
```
print('Median Prevailing wage based on Job experience')
df.groupby('has_job_experience')['prevailing_wage'].median().to_frame().sort_values
```

```
Median Prevailing wage based on Job experience
```

Out[47]:

has_job_experience	prevailing_wage
N	72602.290
Y	69033.665

Report

- Suprisingly people who has less job experience has more prevailing wage.
- No job experience applicant has median prevailing wage as 72k USD.
- Previous job experience applicant has median prevailing wage as 69K USD.

4.8.3 Prevailing wage based on Continent

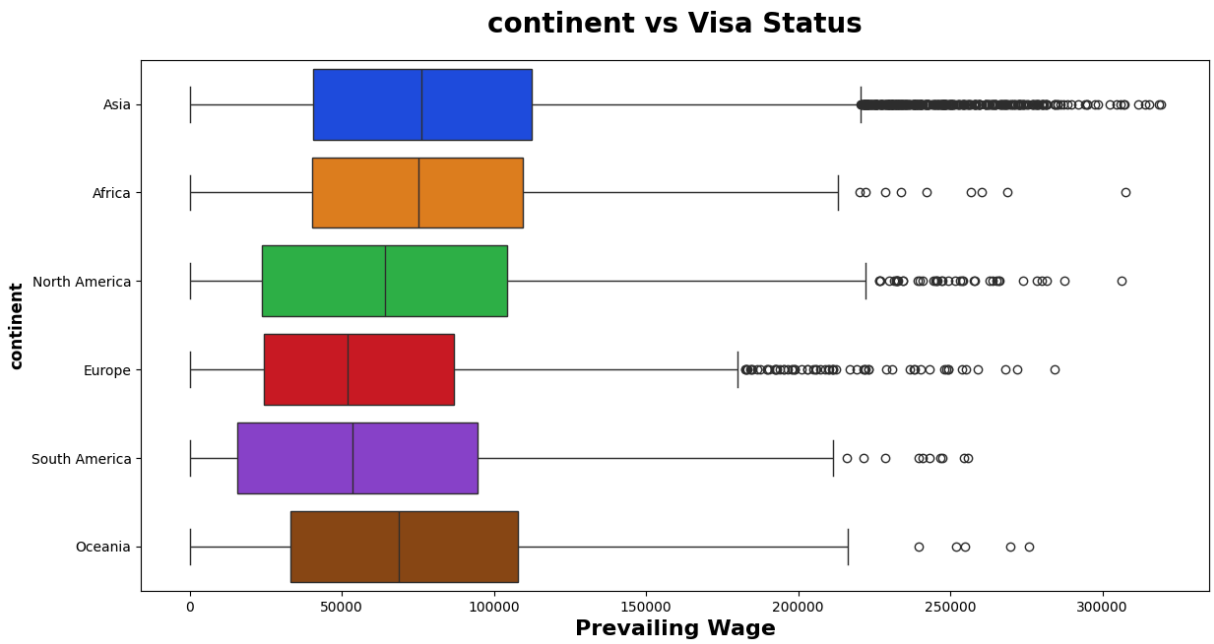
In [48]:
```
print('Average Prevailing wage based on Continent')
df.groupby('continent')['prevailing_wage'].mean().to_frame().sort_values(by='prevai
```

```
Average Prevailing wage based on Continent
```

Out[48]:

continent	prevailing_wage
Asia	79543.021780
Africa	77425.923450
Oceania	75994.276719
North America	68066.319257
South America	60209.575314
Europe	59842.925899

In [49]:
```
plt.subplots(figsize=(14,7))
sns.boxplot(y="continent",x = "prevailing_wage", data=df,palette='bright')
plt.title("continent vs Visa Status", weight="bold",fontsize=20, pad=20)
plt.xlabel("Prevailing Wage", weight="bold", fontsize=16)
plt.ylabel("continent", weight="bold", fontsize=12)
plt.show()
```

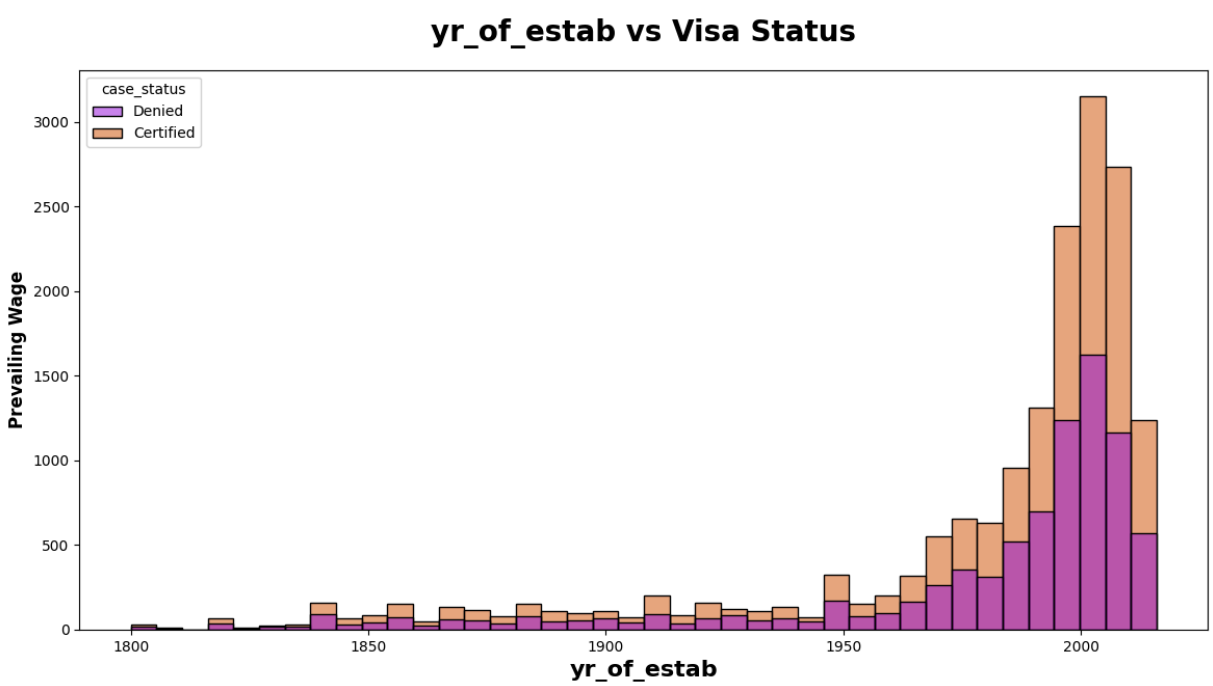


Report

- Average salary of people from Asia is higher than other continents.

4.9 Year of Establishment

In [50]:
```
plt.subplots(figsize=(14,7))
sns.histplot(x = "yr_of_estab", data=df,palette='gnuplot', bins=40, hue='case_statu
plt.title("yr_of_estab vs Visa Status", weight="bold",fontsize=20, pad=20)
plt.xlabel("yr_of_estab", weight="bold", fontsize=16)
plt.ylabel("Prevailing Wage", weight="bold", fontsize=12)
plt.show()
```



Report

- Each bins consists of 5 years.
- Many companies were established after year 2000.
- The most company establishment happened in year 2000-2005.

Final Report

- `case_id` column can be dropped as it is an ID.
- `requires_job_training` column can be dropped as it doesn't have much impact on target variable, Proved in visualization and chi2 test.
- `no_of_employees`, `prevailing_wage` columns have outllier which should be handled.
- `continent` columns has few unique values with very less count, which can be made as **others**
- Target column `case_status` is imbalanced can be handled before model building.

In [ ]:
```
#######################-------------Feature Engineering and Model Training in Next
```